

OPTIMISATION

IN PREDICTING MOTOR TEMPERATURES

PRESENTED BY:

OMAR HANI 13007515

MOHAMED WALID 13006513

HASSAN YOUSEF 13006567

ABDELHAMID ELSHARNOUBY 13006294

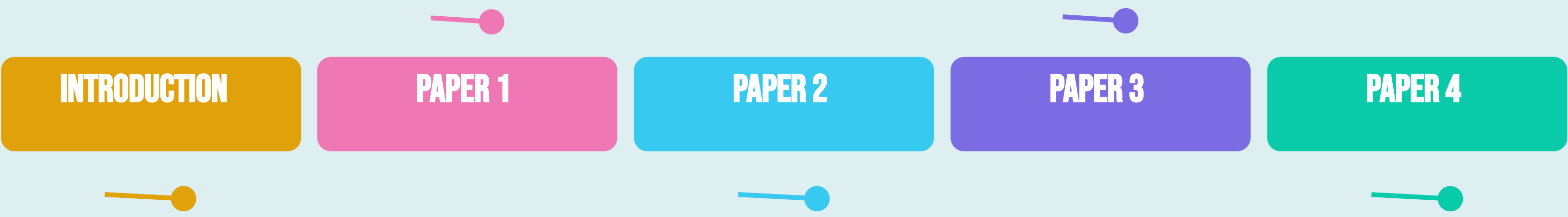
MOHANNAD GHAZAL 13006931

MOHAMED HAMDY 13005467



CONTENTS

WHAT OUR PRESENTATION WOULD BE ABOUT



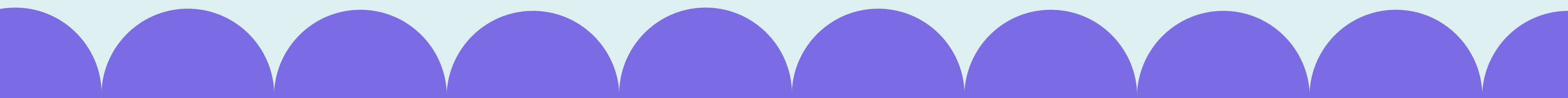
INTRODUCTION

PAPER 1

PAPER 2

PAPER 3

PAPER 4



INTRODUCTION



WHAT IS OPTIMISATION?:

IMPROVING MODEL PERFORMANCE BY FINDING BEST PARAMETERS

USED IN:

TRAINING ML MODELS

HYPERPARAMETER TUNING

REDUCING ERROR (LOSS FUNCTION)

IMPROVEMENT!



WHY OPTIMISATION?

THE USE OF PERMANENT MAGNET SYNCHRONOUS MOTORS (PMSM) TECHNOLOGY IS GROWING IN VARIOUS INDUSTRY APPLICATIONS.

(IN PERMANENT MAGNET SYNCHRONOUS MOTORS)
HOWEVER, PMSMS ARE SUSCEPTIBLE TO HIGH TEMPERATURES ESPECIALLY THE MAGNETS AND WINDINGS.

MACHINE LEARNING OPTIMISATION HELPS BREAKDOWN THE COMPLEX RELATION BETWEEN THE MOTOR'S PARAMETERS INTO NEURAL NETWORKS!

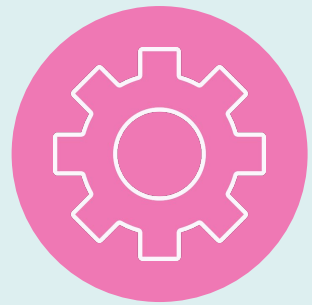
EASIER THAN THERMODYNAMICS!

Estimating electric motor temperatures with deep residual machine learning (Kirchgässner et al, 2021)



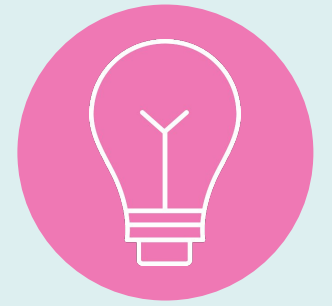
TECHNIQUES

Long Short-Term Memory (LSTM) and
Gated Recurrent Units (GRU);
Bayesian optimisation



CONTRIBUTION

Introduction of deep learning into
thermal monitoring tasks.
No need for physical modeling



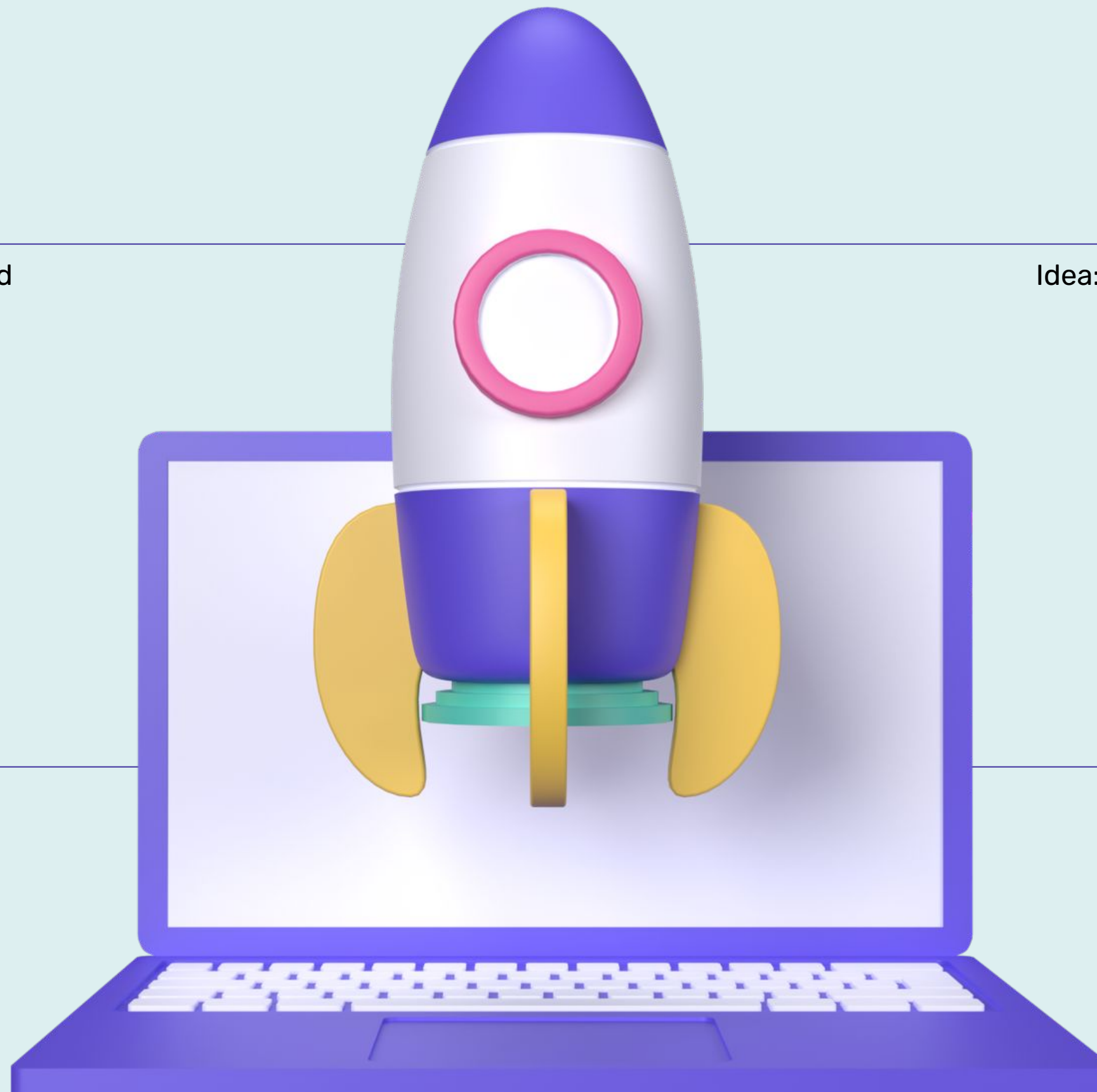
IDEA VS LIMITATIONS

Idea: Combine Deep Learning + Optimisation.
Limitations: Requires large datasets
Hard to interpret (black-box)



RESULTS

MSE below $4K^2$
Maximum absolute deviation $< 9K$



Estimating electric motor temperatures with deep residual machine learning (Kirchgässner et al, 2021)

DATASET? 140 (NOW EXPANDED TO 185) HOURS OF MULTIVARIATE MEASUREMENTS SAMPLED AT 2 HZ FROM A 52 KW PMSM

[WWW.KAGGLE.COM/WKIRGSN/ELECTRIC-MOTOR-TEMPERATURE](https://www.kaggle.com/wkirgsn/electric-motor-temperature) , [HTTPS://GITHUB.COM/WKIRGSN/DEEP-PMSM](https://github.com/wkirgsn/deep-pmsm)

FEATURES?

108 FEATURES INCLUDING AMBIENT TEMPERATURE, COOLANT TEMPERATURE, MOTOR SPEED, DC LINK VOLTAGE, AND CURRENT/VOLTAGE COMPONENTS
EXPANDED VIA EXPONENTIALLY WEIGHTED MOVING AVERAGES (EWMA) AND STANDARD DEVIATIONS (EWMS).

TARGET?

STATOR TEETH, WINDING, YOKE, AND ROTOR PERMANENT MAGNET TEMPERATURES.

ML MODEL?

DEEP RESIDUAL RECURRENT NEURAL NETWORKS (RNNS), TEMPORAL CONVOLUTIONAL NETWORKS (TCNS).

METRICS?

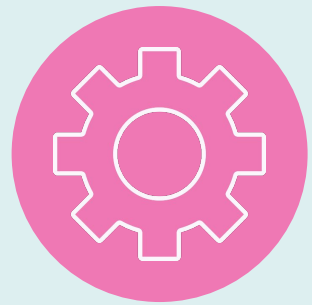
MEAN SQUARED ERROR (MSE) AND MAXIMUM ABSOLUTE DEVIATION (L_{∞} -NORM)

Machine Learning for Sensorless Temperature Estimation of a BLDC Motor (Czerwinski et al, 2021)



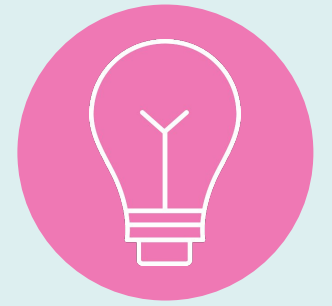
TECHNIQUES

Cross-validation for hyperparameter tuning
Data standardization
Feature engineering using short-term device history



CONTRIBUTION

Proposing a sensorless overheating protection model
Comparing classical ML algorithms



IDEA VS LIMITATIONS

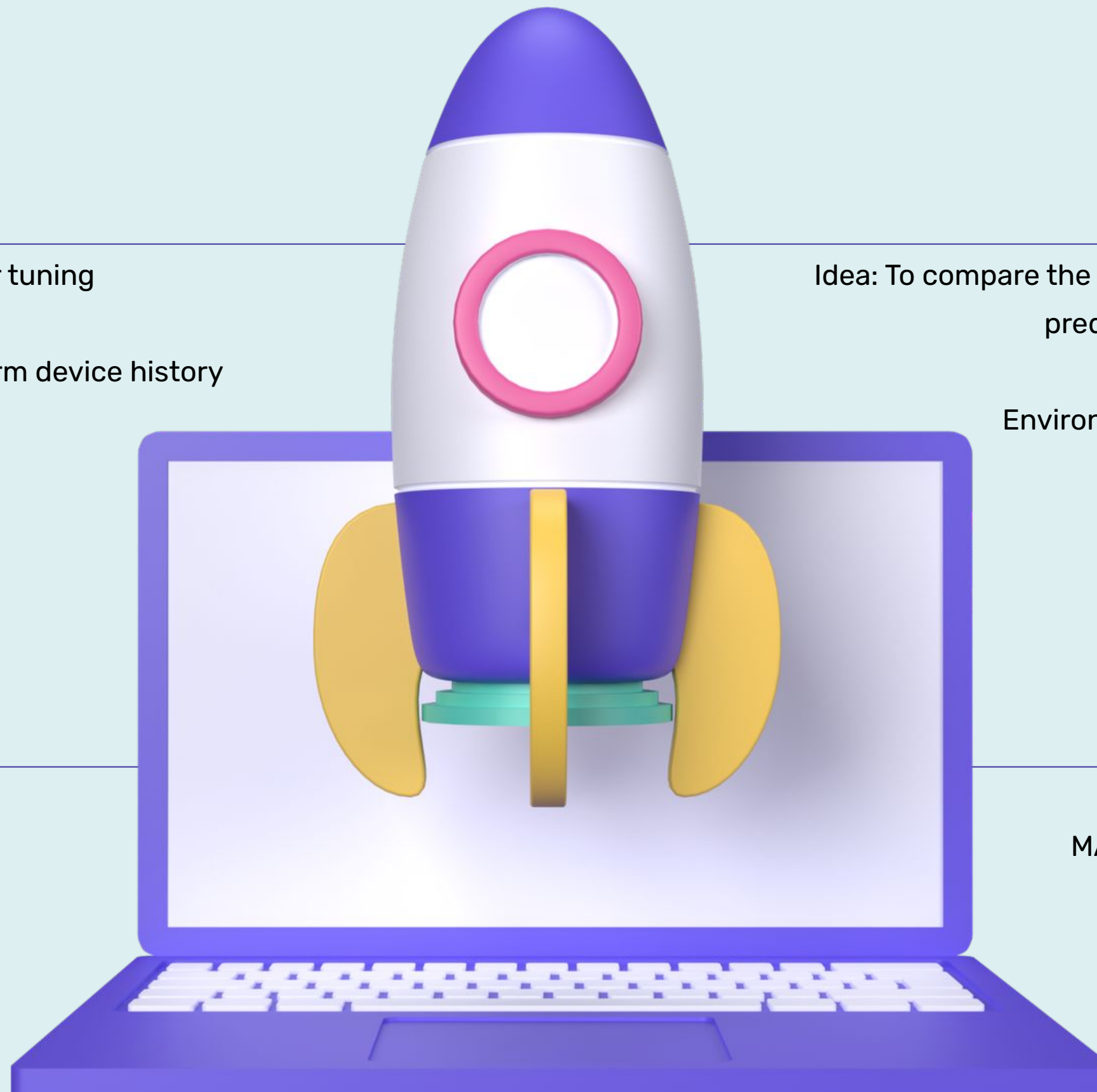
Idea: To compare the effectiveness of various ML algorithms in predicting BLDC motor winding temperature

Limitations: Requires large datasets
Environment sensitivity (ambience and cooling)



RESULTS

Sensorless model (Model 1) achieved a MAPE below 4.5% and an R^2 above 0.909
casing-sensor model (Model 2) reduced MAPE to ~1% with an R^2 of 0.990



Machine Learning for Sensorless Temperature Estimation of a BLDC Motor (Czerwinski et al, 2021)

DATASET? OVER 160 HOURS OF MEASUREMENTS AT 4 HZ FROM TWO COUPLED BLDC MACHINES UNDER VARIABLE LOAD AND SPEED PROFILES.

[HTTPS://IEEE-DATAPORT.ORG/DOCUMENTS/BLDC-MOTOR-TEMPERATURE](https://ieee-dataport.org/documents/bldc-motor-temperature)

FEATURES?

CURRENT, ROTATIONAL SPEED, WINDING POWER LOSSES, AND MEAN AND STANDARD DEVIATION OF CURRENT AND SPEED FROM THE PREVIOUS HOUR.

TARGET?

BLDC STATOR WINDING TEMPERATURE.

ML MODEL?

LINEAR REGRESSION, ELASTICNET, SGD REGRESSOR, SUPPORT VECTOR MACHINES (SVM), DECISION TREES, AND ADABOOST.

METRICS?

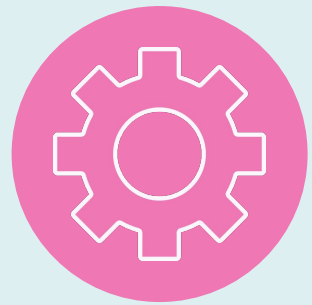
ROOT MEAN SQUARED ERROR (RMSE), MEAN ABSOLUTE PERCENTAGE ERROR (MAPE), MSE, MAX ABSOLUTE DEVIATION, MAE, MRE, AND COEFFICIENT OF DETERMINATION (R^2)

Temperature prediction for electric vehicles of permanent magnet synchronous motor using robust machine learning tools (Al-Gabalawy et al, 2022)



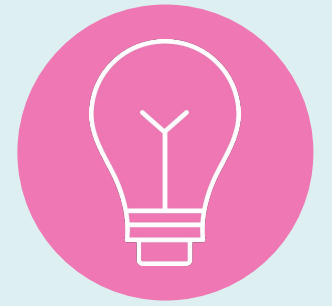
TECHNIQUES

Exploratory data analysis (EDA)
Ridge and Lasso regression
Polynomial regression
Tenfold cross validation



CONTRIBUTION

Predicts PMSM temperature
Detailed feature importance matrix
No thermodynamic modelling needed



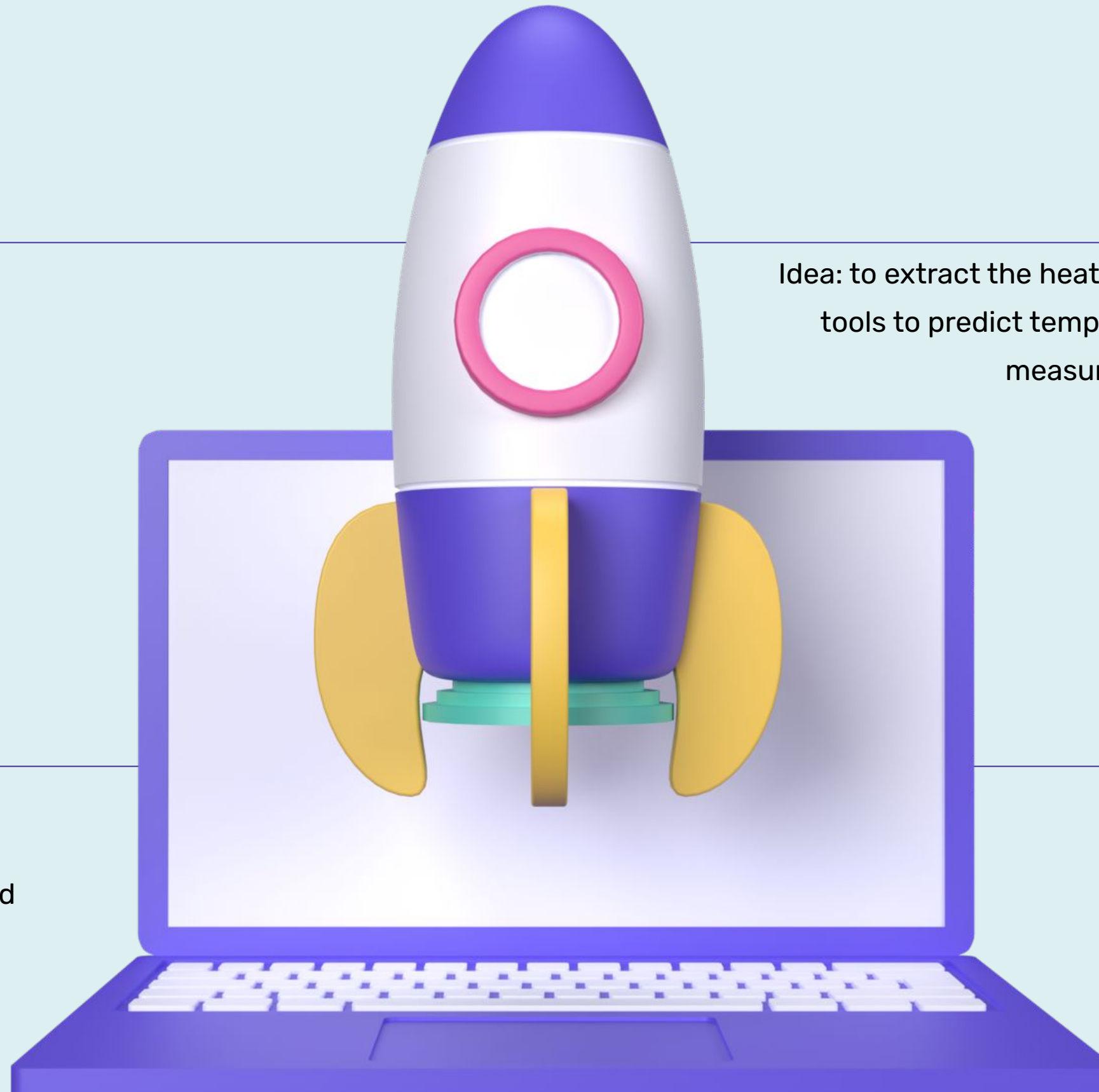
IDEA VS LIMITATIONS

Idea: to extract the heat of vital EV motor parts (PMSM) using ML tools to predict temperatures from specifications and current measurements when sensors are hard to install
Limitations: data-dependency



RESULTS

SVM was the best model with a Test RMSE of 0.59 standard deviations
 R^2 values for polynomial regression reached between 0.593 and 0.682



Temperature prediction for electric vehicles of permanent magnet synchronous motor using robust machine learning tools (Al-Gabalawy et al, 2022)

DATASET? USES A DATASET WITH SPECIFIC IDENTIFIERS THAT ARE IDENTICAL TO THE PUBLIC DATASET RELEASED BY KIRCHGÄSSNER ET AL.

15,147 INSTANCES WITH 13 ATTRIBUTES.

FEATURES?

AMBIENT TEMPERATURE, COOLANT TEMPERATURE, VOLTAGE COMPONENTS, MOTOR SPEED, MOTOR TORQUE, AND CURRENT COMPONENTS.

TARGET?

ROTOR TEMPERATURE (PM)

ML MODEL?

EXTRA-TREE, BAGGING K-NEAREST NEIGHBORS (KNN), VOTING REGRESSOR, RANDOM FOREST, SVM, AND XGBOOST.

METRICS?

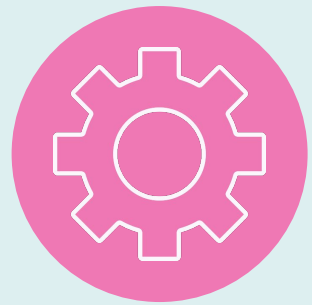
ROOT MEAN SQUARED ERROR (RMSE) AND COEFFICIENT OF DETERMINATION (R^2)

Thermal neural networks: Lumped-parameter thermal modeling with state-space machine learning (Kirchgässner et al, 2023)



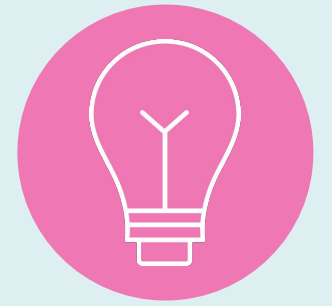
TECHNIQUES

First-order Euler discretization, state-space representation, and kernel-based hyperparameter optimization (HPO)



CONTRIBUTION

Proposes the TNN approach which solves the limitations of previous black-box (RNN) and expertise dependent (LPTN) approaches by joining them together.



IDEA VS LIMITATIONS

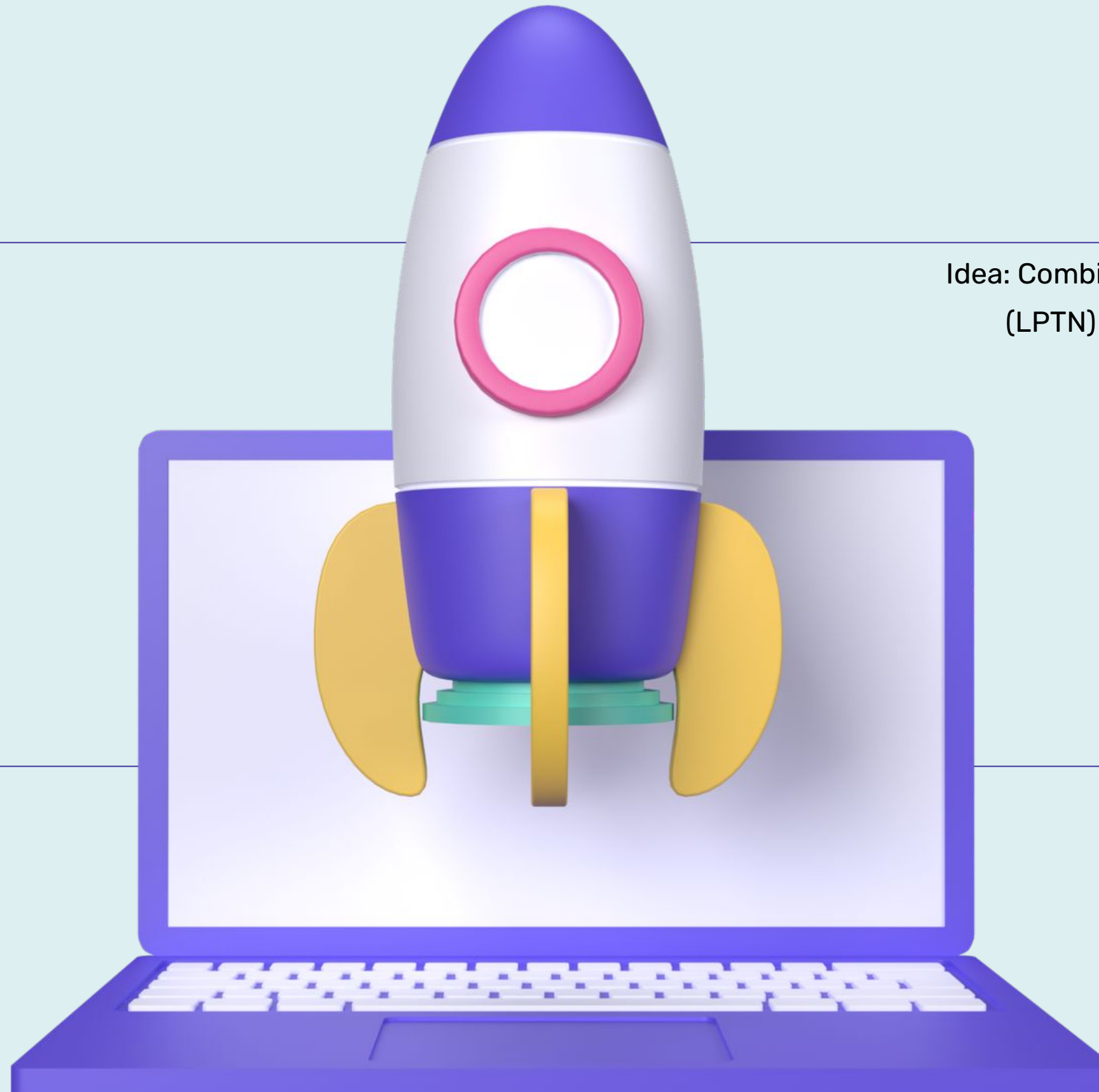
Idea: Combine lumped-parameter thermal networks (LPTN) + state-space machine learning (SSNN).

Limitations: Requires large datasets



RESULTS

Achieved an MSE of $3.18K^2$
Maximum absolute deviation $< 5.84 K$



Thermal neural networks: Lumped-parameter thermal modeling with state-space machine learning

(Kirchgässner et al, 2023)

DATASET? 185 HOURS OF MULTIVARIATE MEASUREMENTS SAMPLED AT 2 HZ FROM A 52 KW PMSM

[WWW.KAGGLE.COM/WKIRGSN/ELECTRIC-MOTOR-TEMPERATURE](https://www.kaggle.com/wkirgsn/electric-motor-temperature) , [HTTPS://GITHUB.COM/WKIRGSN/THERMAL-NN](https://github.com/wkirgsn/thermal-nn)

FEATURES?

VOLTAGE VECTOR NORM, CURRENT VECTOR NORM, MECHANICAL ANGULAR FREQUENCY AND AMBIENT AND COOLANT TEMPERATURES.

TARGET?

STATOR TEETH, WINDING, YOKE, AND ROTOR PERMANENT MAGNET TEMPERATURES.

ML MODEL?

HYBRID TNN.

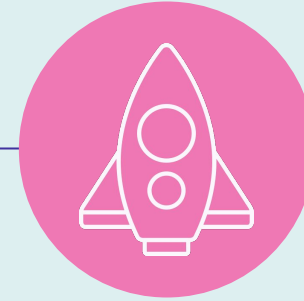
METRICS?

MEAN SQUARED ERROR (MSE) AND MAXIMUM ABSOLUTE DEVIATION (L_{∞} -NORM)

COMPARISON TABLE

PAPER	MODELS AND TECHNIQUES	OPTIMISATION TYPE	PERFORMANCE METRICS AND RESULTS	STRENGTH	LIMITATION
[KIRCHGASSE ET AL, 2021]	RNN, TCN, LSTM, AND GRU.	BAYESIAN OPTIMISATION ADAM OPTIMISER	MSE < 4K ² MAXIMUM ABSOLUTE DEVIATION < 9K	HIGH ESTIMATION ACCURACY NO REQUIRED EXPERTISE REAL-TIME ADAPTABILITY	BLACK-BOX DATA DEPENDENCY
[CZERWINSKI ET AL, 2021]	LINEAR REGRESSION, ELASTICNET, SGD REGRESSOR, SVM, DECISION TREES, AND ADABOOST.	EXHAUSTIVE GRID SEARCH ADAM/SDG OPTIMISER	MAPE <4.5% AND R ² > 0.909 (MODEL 1), MAPE OF 1% AND R ² OF 0.99 (MODEL 2) (AND MORE)	HIGH ESTIMATION ACCURACY	DATA DEPENDENCY
[AL-GABALAWY ET AL, 2022]	KNN, VOTING REGRESSOR, RANDOM FOREST, SVM, XGBOOST, AND EDA.	TENFOLD CROSS-VALIDATION RIDGE AND LASSO REGRESSION	RMSE OF 0.59 STANDARD DEVIATION (SVM)	HIGH ESTIMATION ACCURACY NO REQUIRED EXPERTISE REAL-TIME ADAPTABILITY	DATA DEPENDENCY ENVIRONMENT SENSITIVITY
[KIRCHGASSE ET AL, 2023]	TNN (LPTN + SSNN).	KERNEL-BASED HPO NADAM OPTIMIZER	MSE OF 3.18 K ² MAXIMUM ABSOLUTE DEVIATION < 5.84 K	HIGH ESTIMATION ACCURACY NO REQUIRED EXPERTISE REAL-TIME ADAPTABILITY	DATA DEPENDENCY

BEST SOURCE



EFFICIENCY



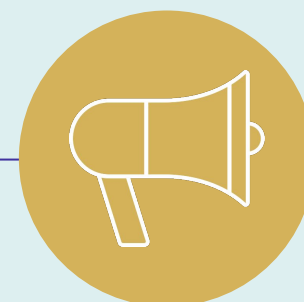
Achieves this higher accuracy with fewer parameters (64 parameters compared to thousands in black-box models).



ACCURACY



It achieved a Mean Squared Error (MSE) of $3.18 K^2$ and a worst-case error (L_∞ -norm) of $5.84 K$



COMPARISON



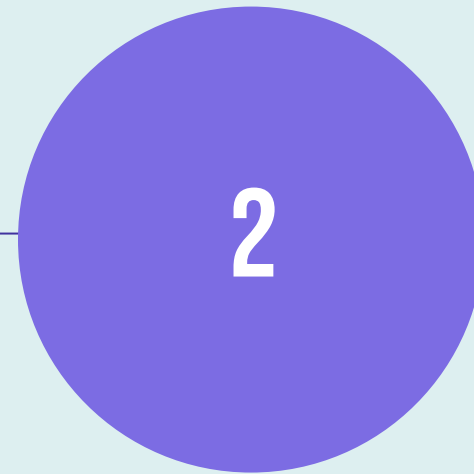
Higher Accuracy than other sensorless models in papers 1, 2 and 3. Although, source 2 shows high accuracy with casing sensors.

3 IDEAS! (AND WE CHOOSE THE FIRST ONE)



EXPAND SOURCE 1

(Kirchgässner et al, 2021) originally used a 140-hour dataset. (Kirchgässner et al, 2023), utilizes an expanded 185-hour dataset that is already publicly available on Kaggle. By using the extra 45 hours of data, we can verify whether the performance keeps improving.



IMPROVE SOURCE 2

Applying a TNN to the BLDC motor would solve several problems mentioned in (Czerwinski et al, 2021): it would provide physically interpretable states, allow for initial condition adjustment, and likely achieve higher accuracy with far fewer parameters than the classical models



APPLY SOURCE 4

(Kirchgässner et al, 2023) suggests that the TNN and state-space machine learning principles should be applied to other domains like battery systems (state of charge) or power electronic converters



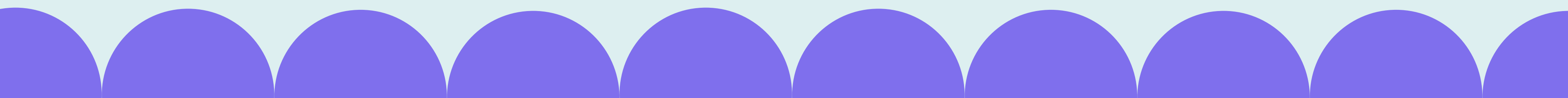
References

W. Kirchgässner, O. Wallscheid, and J. Böcker, "Estimating Electric Motor Temperatures With Deep Residual Machine Learning," IEEE Transactions on Power Electronics, vol. 36, no. 7, pp. 7480-7488, July 2021.

D. Czerwinski, J. Gęca, and K. Kolano, "Machine Learning for Sensorless Temperature Estimation of a BLDC Motor," Sensors, vol. 21, no. 14, p. 4655, July 2021.

M. Al-Gabalawy, A. H. El Metwally, R. A. Younis, and A. I. Omar, "Temperature prediction for electric vehicles of permanent magnet synchronous motor using robust machine learning tools," Journal of Ambient Intelligence and Humanized Computing, vol. 15, pp. 243-260, May 2022.

W. Kirchgässner, O. Wallscheid, and J. Böcker, "Thermal neural networks: Lumped-parameter thermal modeling with state-space machine learning," Engineering Applications of Artificial Intelligence, vol. 117, p. 105537, 2023.



THANK YOU!

HAVE A “COOL” DAY!

